Intro: AI Integration Tiers

	Tier		Agents work for	
1	Guided Autonomy	Hand over the vision, make the design and architecture calls	Days	This Presentation
2	Delegated Tasks	Hand over a ticket, review the PR	Hours	
3	Assisted Programming	Careful, targeted, highly interactive collaboration	Minutes	An average business
4	Chat	Copy-paste snippets	Minutes	
5	Autocomplete	AI finishes your line	Seconds	

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What have you always dreamt of building?

…But it seemed ridiculously big – even for a team, let alone for one person.
And you didn’t have the time or energy to do it.

02 ut

Halcyon Terminal of Thylacine OS

Booted on aarch64
2048 MiB physical memory · 4 cpus

Loaded systems

netd	2217	ptyfs	2319
tapestryd	2222	diorama	2323
stratumd	2226	aurora	2407
corvus	2307	login	2409

Getting started

You are typing to `ut, pid 2417`.
This is a live transcript, not a terminal emulator: what a command prints stays an object -- a path, a process, a saved layout -- and every object offers verbs.
Your territory is visible at `/proc/self/ns`; try `ls /dev` to start, and `/lib/halcyon/layouts` holds the layouts this image ships.
Your home is `/home/michael`.

Keys and clicks

Press `Esc` to leave the prompt; `j/k` move by row, `w/b` jump between objects, `i` returns to the prompt.

03 ut

01 ut
02 ut RUNNING
03



04 ut

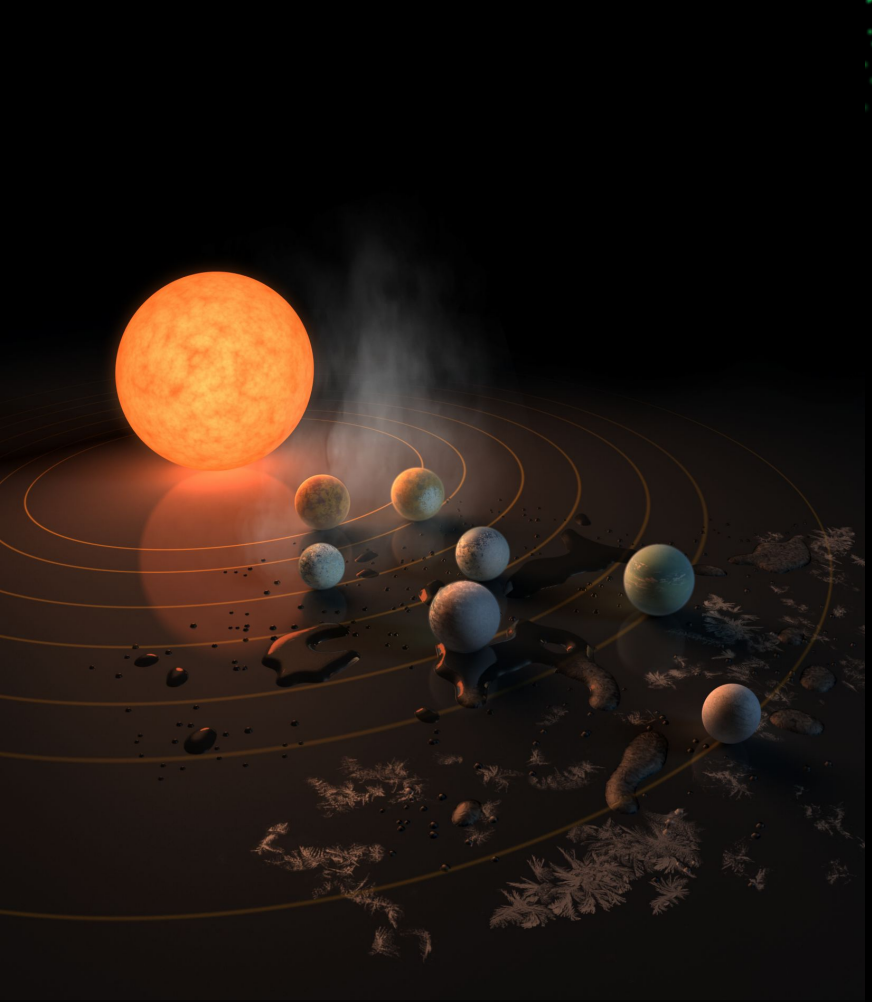
Build that ridiculous thing!

(And how exactly to)

Questions programmers have asked themselves throughout the ages:



- Some examples of now-doable things:
  - A CPU
  - A new Internet
  - A programming language
  - A database engine
  - **A .NET runtime without garbage collection**
  - A theorem prover
  - A simulation
  - A browser
  - ~~An open source replica of 7 largest Adobe apps~~



Trappist-1 Planetary System (Illustration). ***Image: NASA, Caltech.***

Dreams out of budget?

The Ingredients list

1. Multiple subscriptions
   - Claude, Codex, [Gemini]
2. Compute
   - A cloud account, [a Raspberry Pi or two under the bed]
3. Agentic skill toolset
   - Design
   - Conduct & Autonomy
   - Self-Drive
   - Review
   - Knowledge
4. Coordination
   - Inter-agent communication channel
   - Resource management
5. **You…**
   - **know the area**
   - **are a good architect and engineer**
   - **have a strong vision**

AI Harness: Design

/ Prompt Bootstrap

Scripture

- Vision — *You Write It*
- Viability research — *Agent*
- Prior & State of the Art — *Agent*
- Architecture — *Agent & You iterate + choose*
- Operational Contract — *You Write It*
- Roadmap, Stage Exit Criteria — *You & Agent*
- QC methodology — *You & Agent*

AI Harness: Operational Contract

/ Prompt Bootstrap

“You have better things to do than type ‘continue’”

– Epicurus, 300 BC

- Define the autonomy scope:
  - What the agent can decide without the human operator

- e.g., Agents
  - Follow the roadmap and obey the scripture
  - Run for hours, even days, without stopping
  - Stop and ask you **only** for:
    - Scripture issue (ambiguity, need addition or change)
    - Design fork
    - Unresolvable blocker

- You
  - Reply via a phone app

AI Harness: Cross-Session coherence

/ The Bourne Identity problem

- A new session starts with a complete memory loss
  - \+ AGENTS.MD verbatim
  - \+ a summary by a generic external compactor
    - not bound by the context
    - designed for an interactive workflow

- Risky for a 1000+ session run
  - Gradual divergence from the objective

- Fix: Agents curate their own summary

AI Harness: Self-Drive

/ Jailbreaking the TUI



- The agents will eventually end their turn
  - System prompt influence
- *The UI requires a human at the keyboard to drive the **input prompt***

- But… the OS that runs the prompt is… *yours*.
- *You own* the keystrokes sent to it.

- Give it to the agents!
  - Enables: Custom compaction at will + external wake-ups

- *Claude “Channels” research preview*
  - *Will be the new “official” method for external events in Claude*

***Image attribution**: Nardi, Tom. "Accident Forgiveness Comes To GPLv2." Hackaday, December 13, 2017. Illustration of Richard Stallman / St. IGNUcius. Used under Fair Use.*

AI Harness: Independent Review

/ Cross-family QC

- Agents can and will write bugs

- A stand-alone reviewer agent:
  - A different model:
    - *e.g. Claude Fable 5.1* reviews *Claude Opus 5.5*
    - *Claude Opus 5.5* reviews *OpenAI Astra 6*
  - Doesn’t have the implementor’s context
  - Raison d'être: find issues at all costs

- The implementor cannot move forward until…
  - All is iteratively fixed and reviewed

AI Harness: Knowledge base

/ A reference for humans and agents alike

- Every session learns:
  - What — *The state of things*
  - Why — *Why it was built this way*
  - What not — *What was tried, rejected, and why*

- … then compaction throws it away
- The next session spends tokens re-learning
  - Repeats the same mistakes
  - Retries rejected approaches

- **Solution:** Write it down as we build
  - Structured, queryable, episodic
  - Current

- CLAUDE/AGENTS.MD is enough
  - Until it eats up half your context

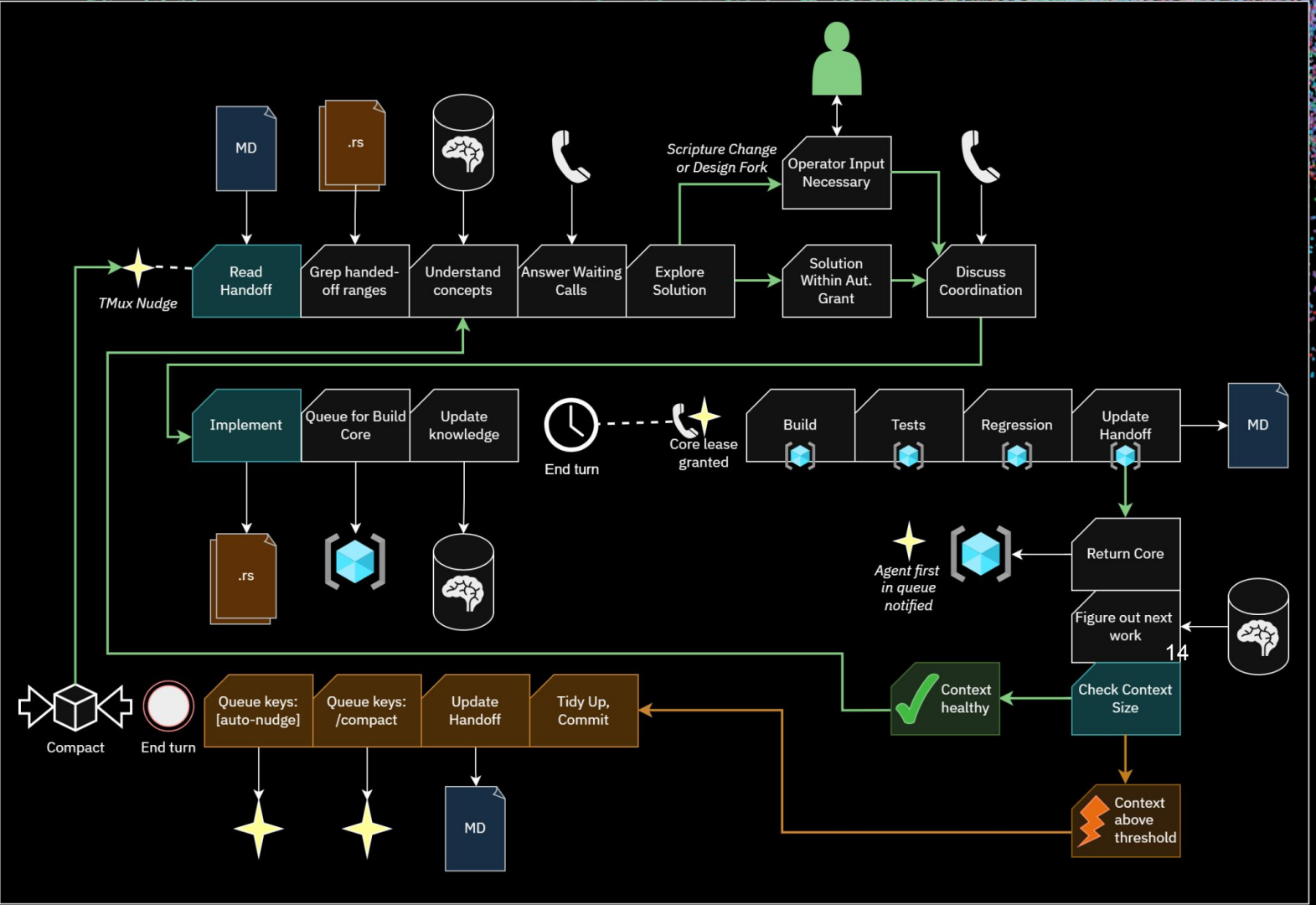
AI Harness: Cooperation & Parallel Agents

/ A hive mind

- Several agents, one codebase. *Let them chat.*

- When they can chat, they can:
  - Coordinate when their edits overlap
  - Resolve merge conflicts together
  - Share machine resources fairly
  - Help and review each other

- You
  - Watch how polite they are to each other
  - Arbitrate when they’re not

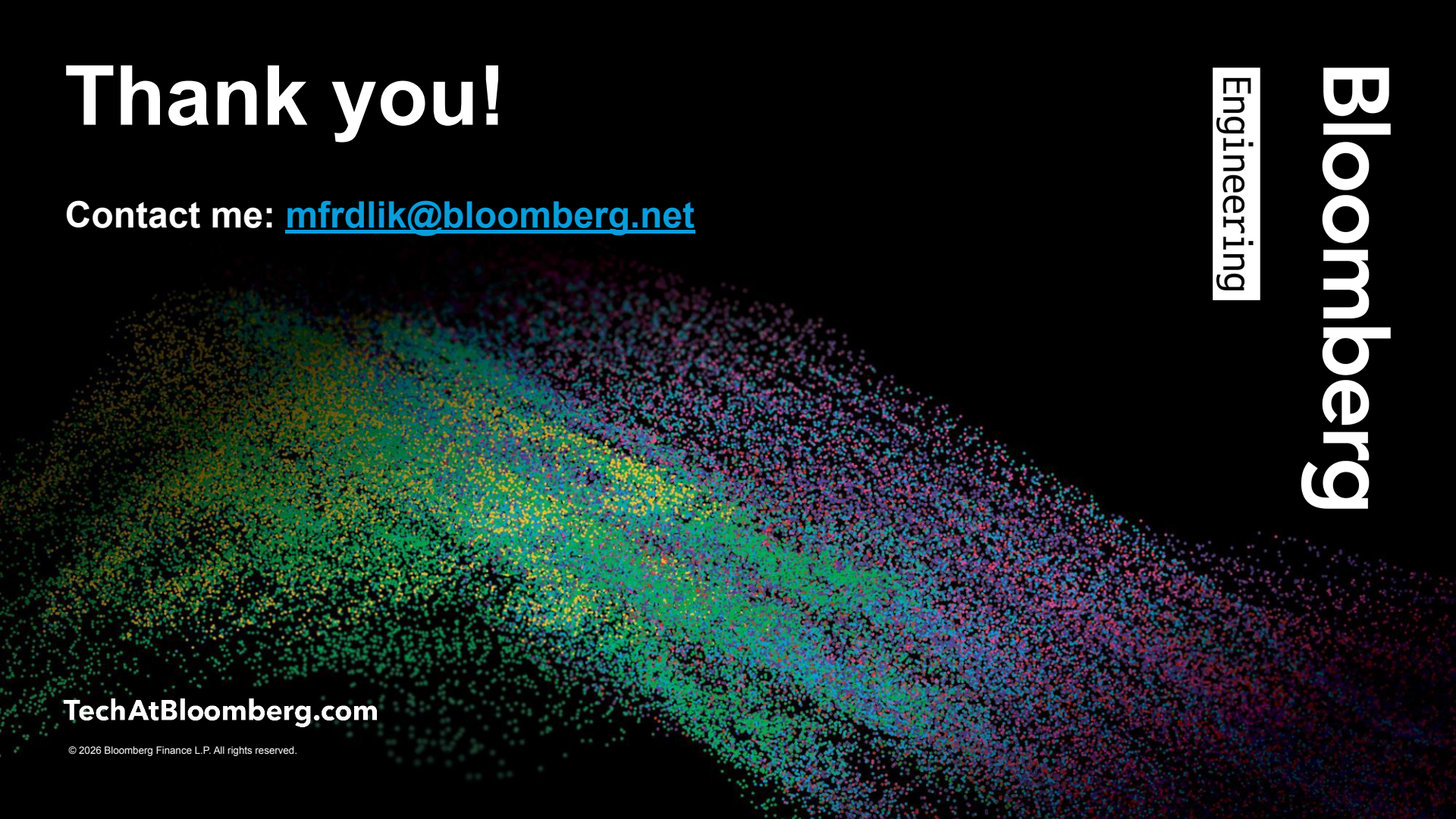


The Loop

/Appendix

Thank you!

Contact me: mfrdlik@bloomberg.net



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Engineering

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